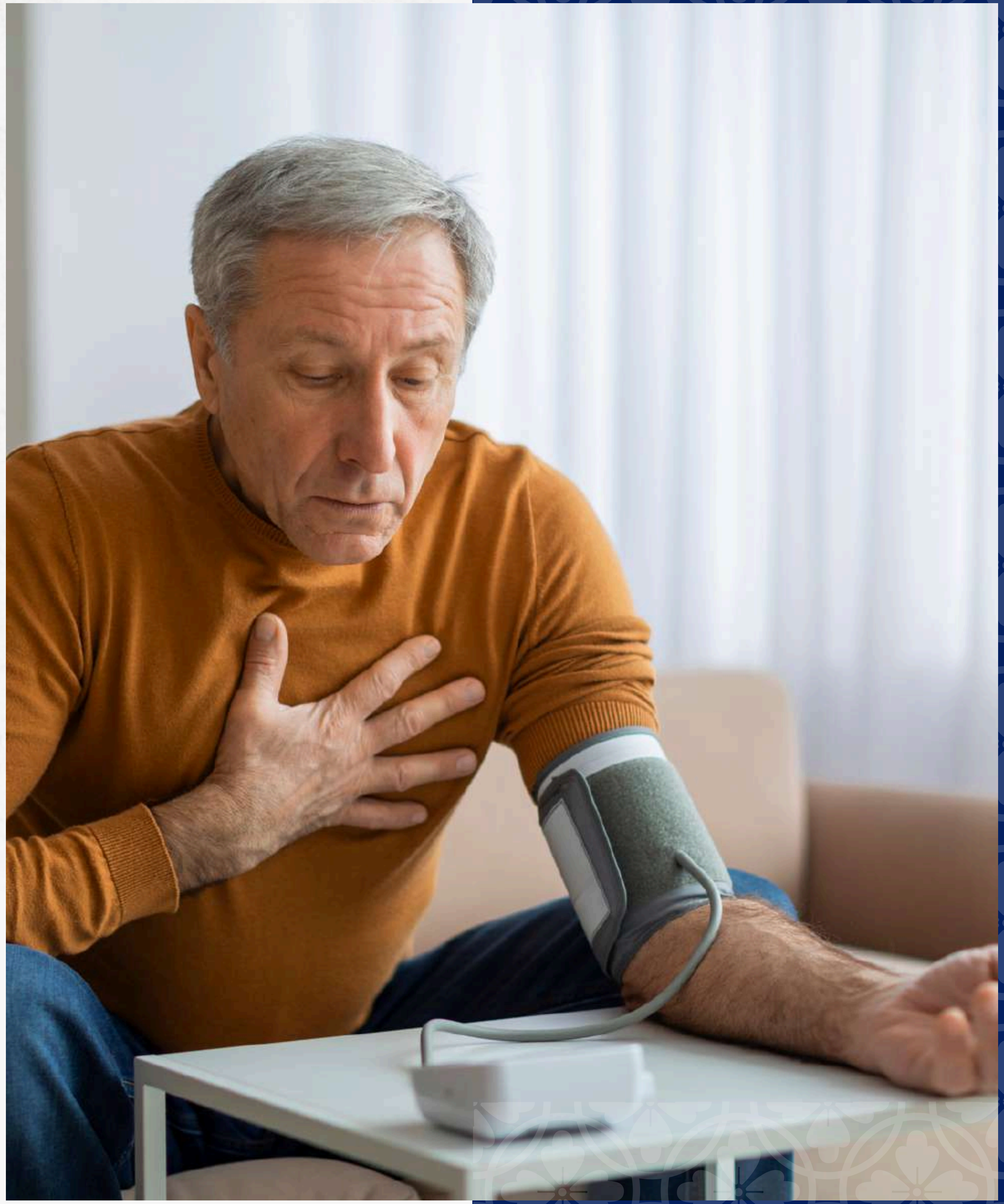
CPE391 Problem-Based Learning

NCDS

NON-COMMUNICABLE DISEASES PREDICTION

This aims to predict the risk of developing non-communicable diseases such as diabetes among older adults by utilizing personal information, lifestyle behaviors, and health examination data.




CONCEPTUAL DESIGN

Conceptual Design

MOVING TARGET

User

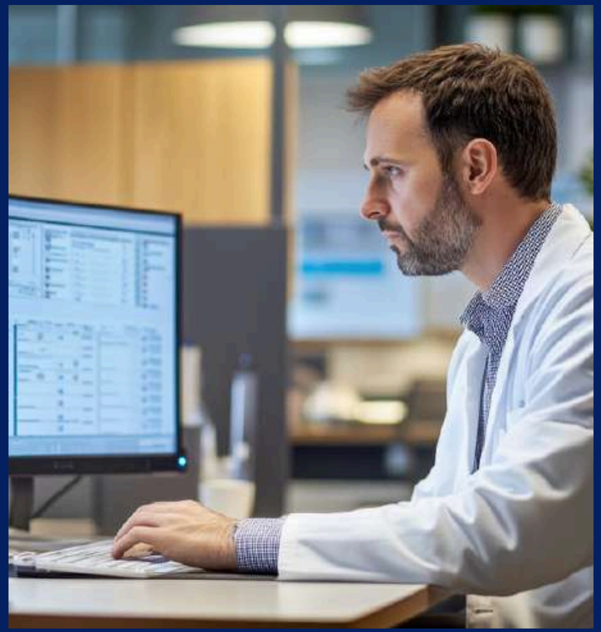
Stakeholders

Main

External

Related hospital personnel

Hospital administrators



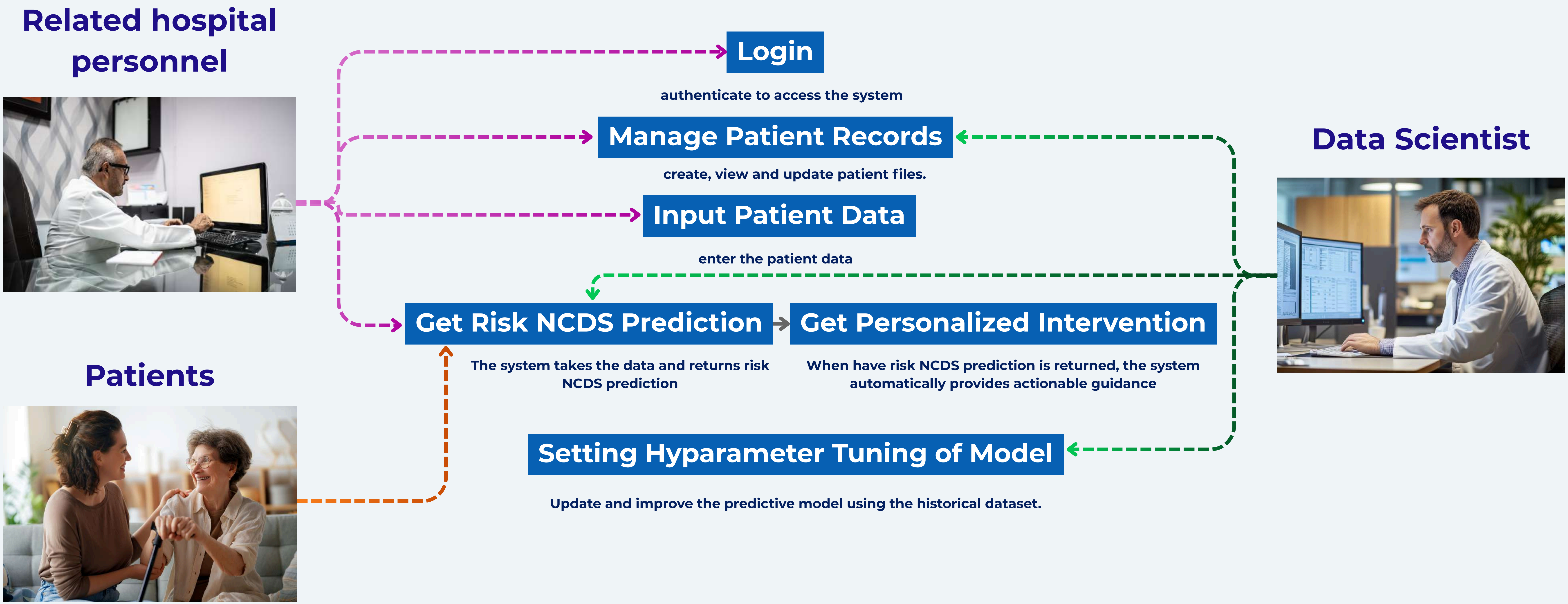
Data Scientist

Patients

Patient's family

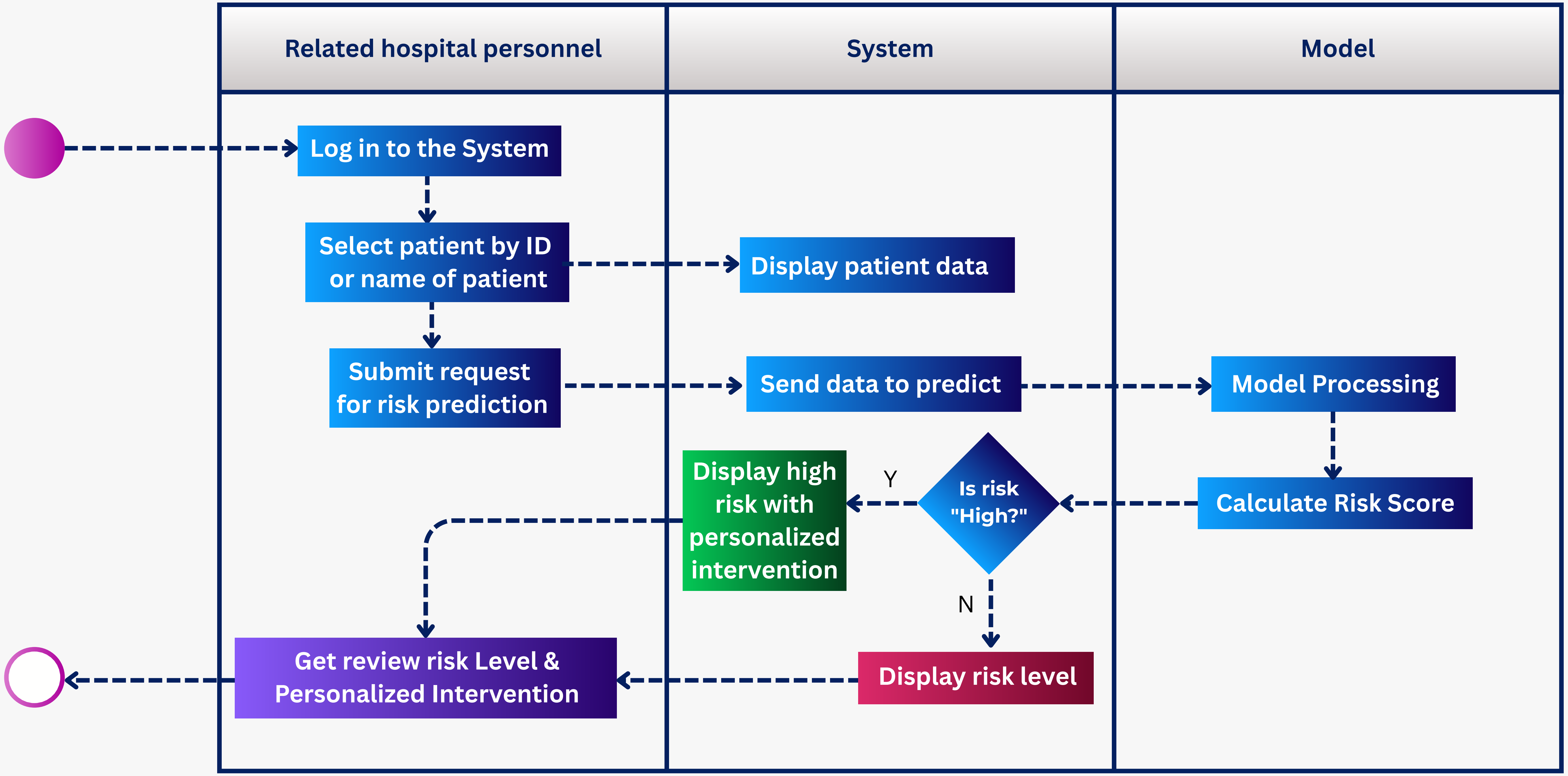
Conceptual Design

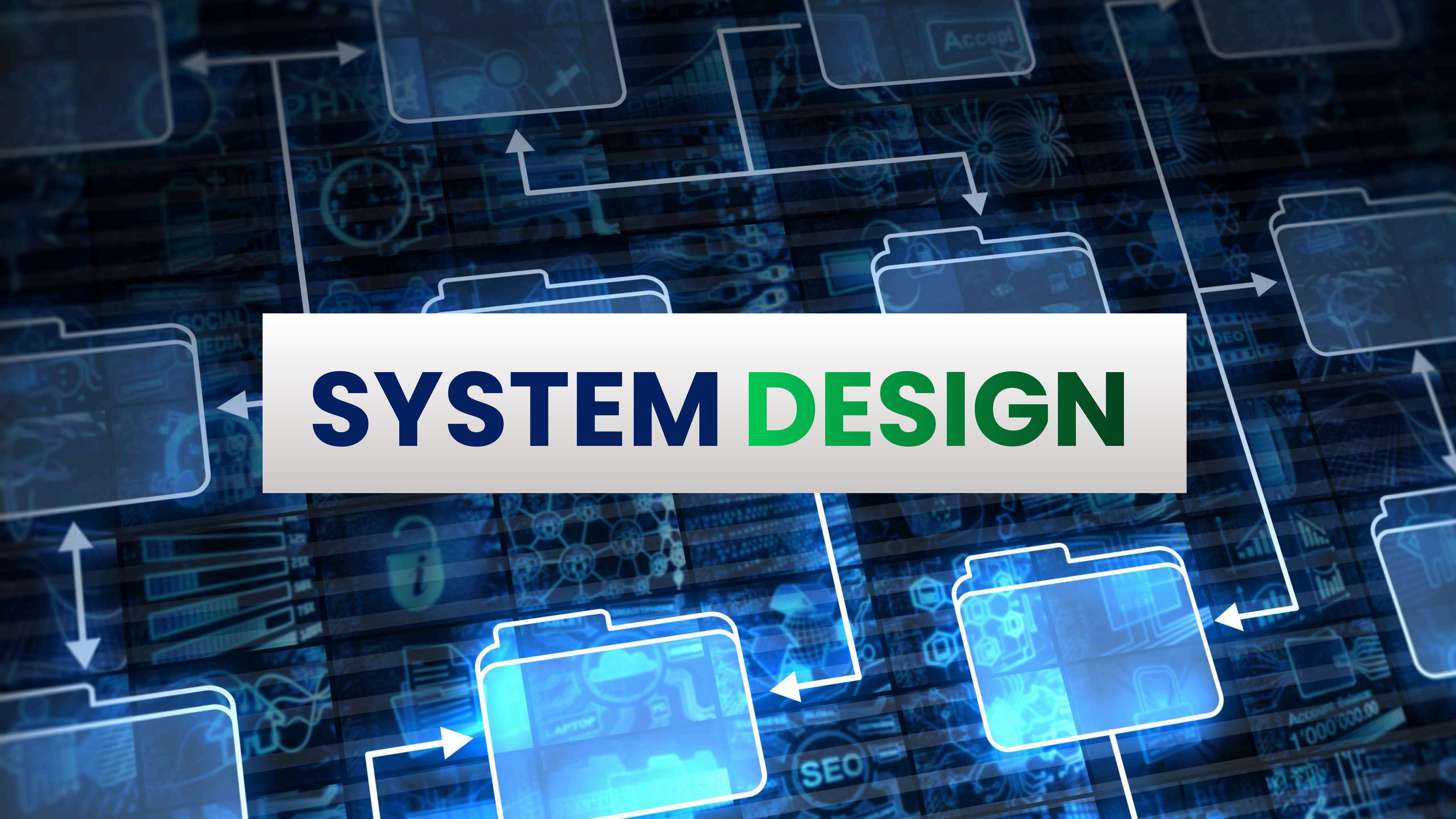
USE CASE DIAGRAM



Conceptual Design

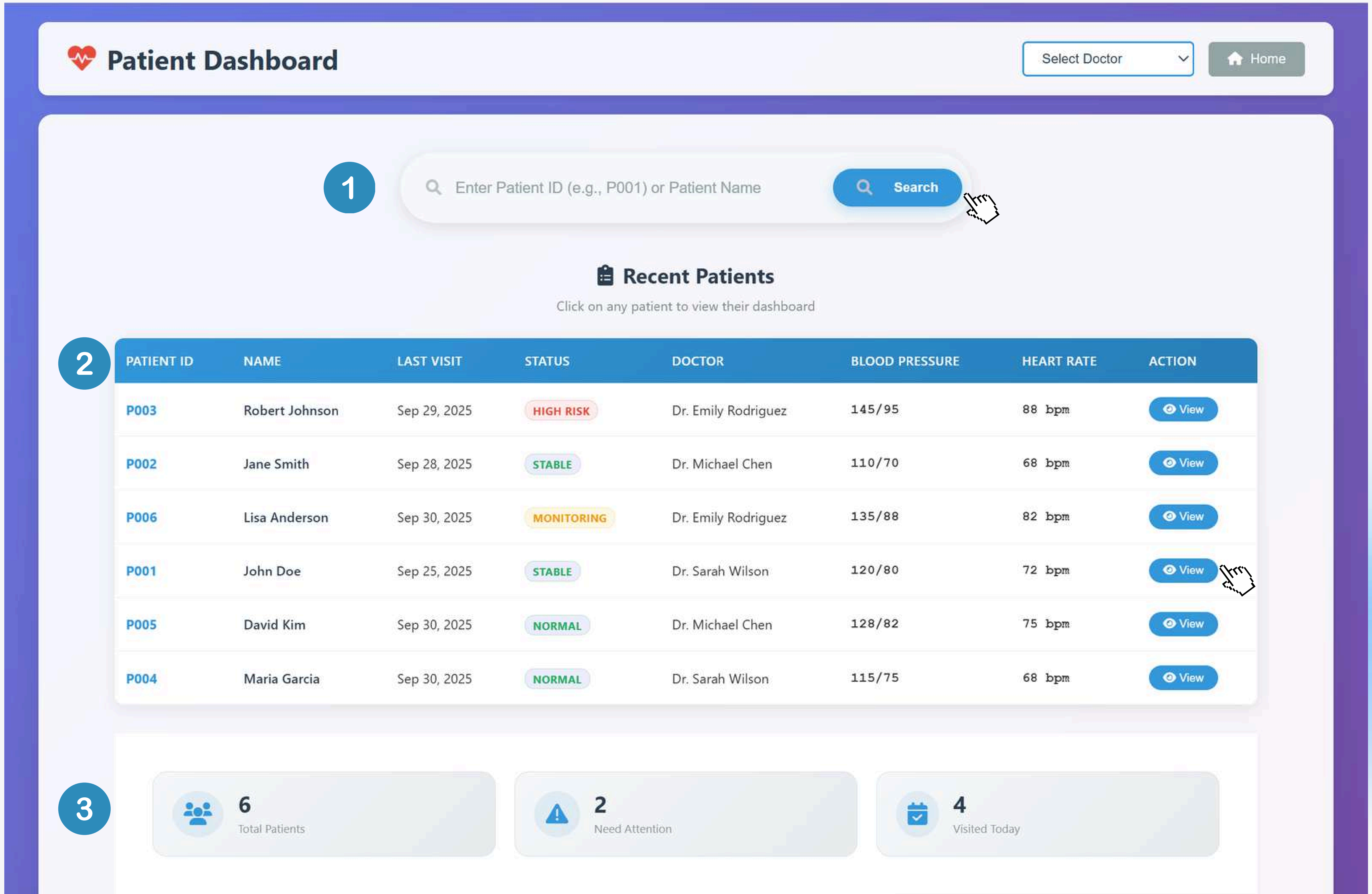
ACTIVITY DIAGRAM




SYSTEM DESIGN
Accept
SEO

Specification Design

UI FUNCTIONS & SYSTEM SEQUENCE



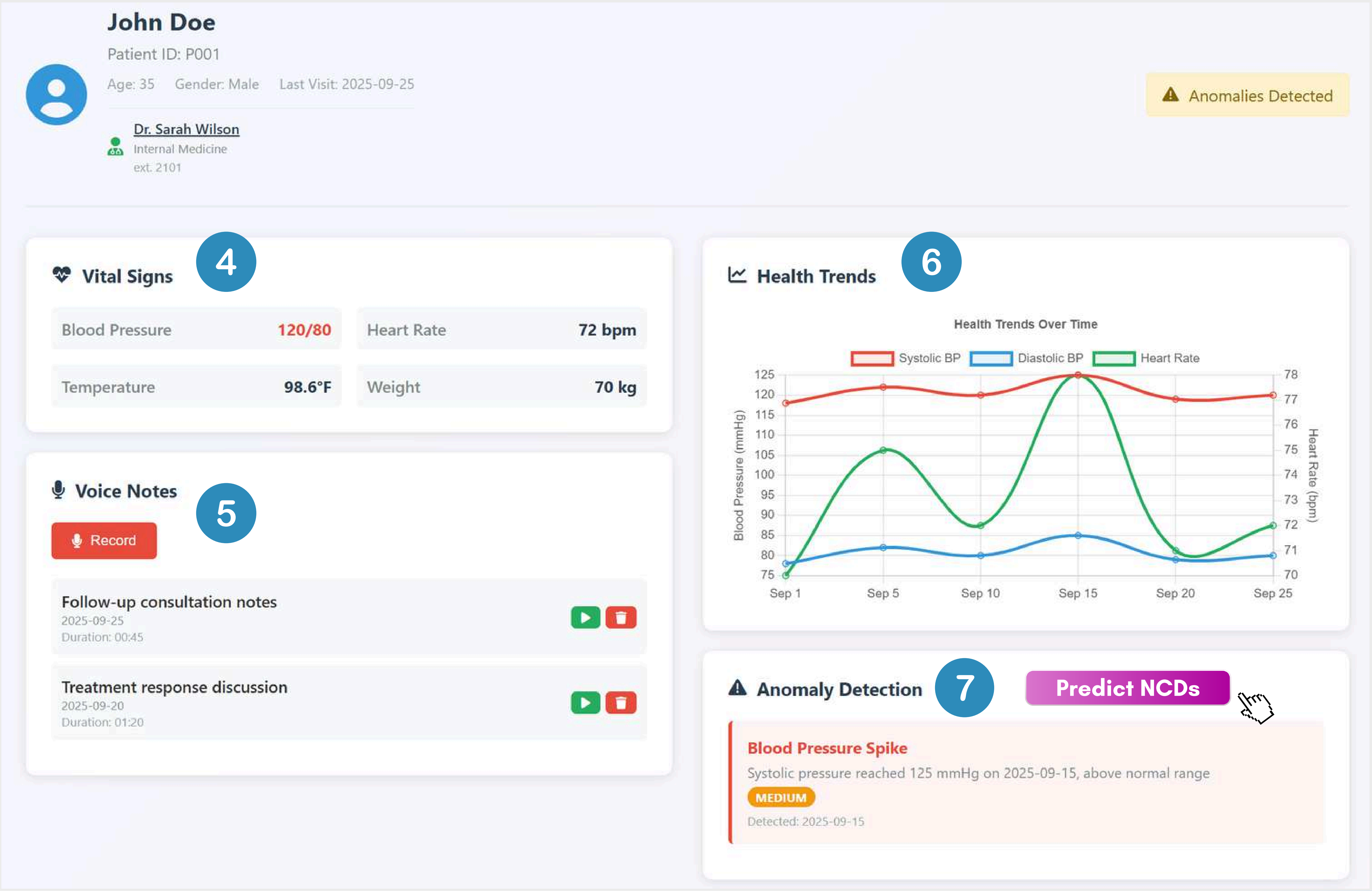
1. **Patient Search**
   by patient ID or name of patient

2. **Patient Table**
   show last visit, status, doctor and infomation of heath

3. **Patient Dashboard**
   show total patient, NCDs attention and amount visited today

Specification Design

UI FUNCTIONS & SYSTEM SEQUENCE



4 **Vital Signs**

show blood pressure, heart rate, temperature and weight

5 **Voice Notes Record**

record result of treatment by doctor's voice

6 **Health Trends**

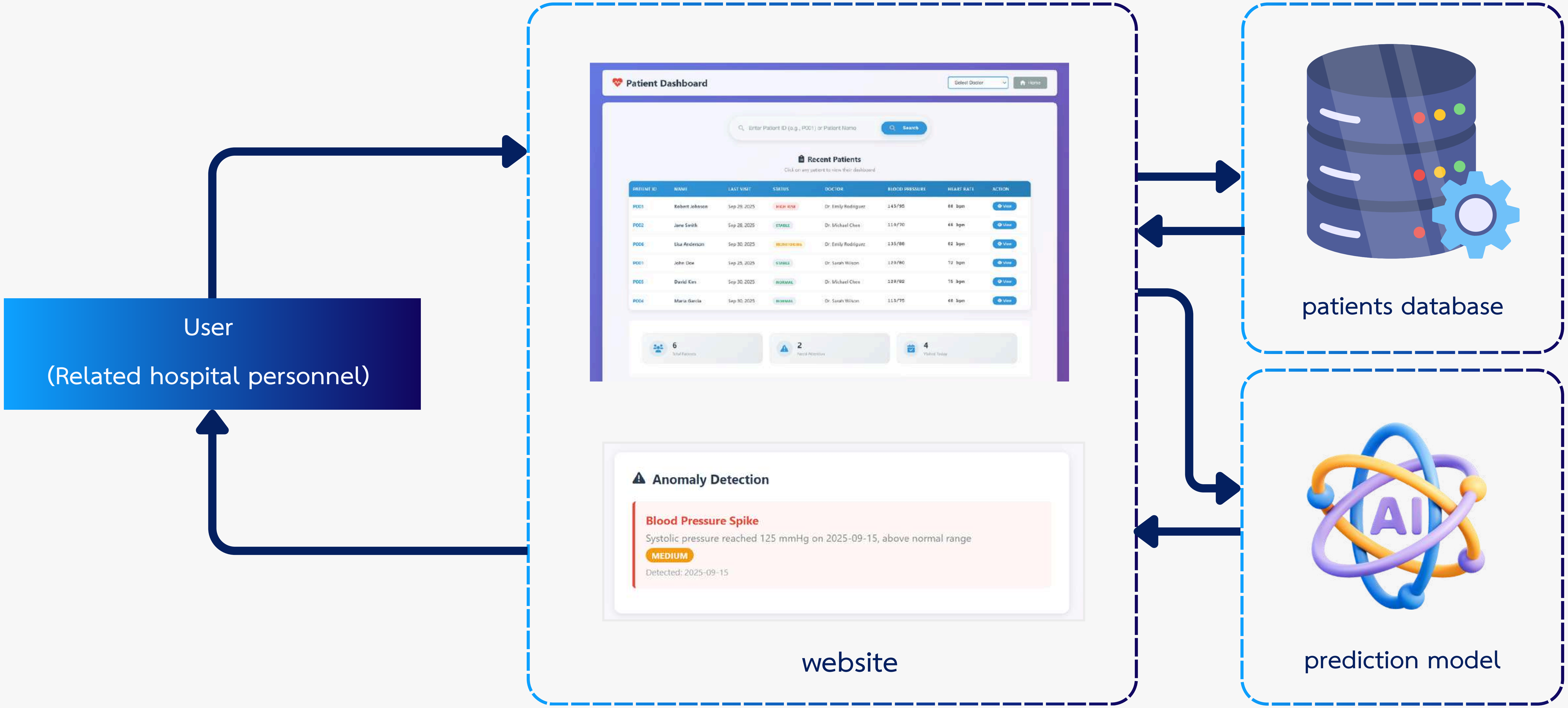
show heath trends such as blood pressure

7 **Anomaly Detection**

show risk of NCDs

Specification Design

SYSTEM CONFIGURATION



Conceptual Design

DATABASE DIAGRAM

1

PatientInfo
SubjectID Varchar (17) PK
Gender Varchar (5)
Birthday Date
Age Integer
Height Float
BodyWeight Float
Disease name Varchar (30)
Smoking Boolean
Drinking Boolean
Activities Boolean

Master table

*

Lifestyle
SubjectID Varchar (17) FK
Appetite Varchar (8)
EatingHabit Varchar (8)
Preference Varchar (8)
AnxietyHealth Varchar (8)

*

InspectionResult
SubjectID Varchar (17) FK
InspectionDate Date
BloodSugar Float
HbA1c Float
TotalCholesterol Float
LDL Float
HDL Float
Triglyceride Float
Creatinine Float
DiabetesDiagnosisAge Float

*

Evaluation
SubjectID Varchar (17) FK
HealthStatus Varchar (12)
BiologicalFunction Varchar (12)
ExerciseFunction Varchar (12)
Nutrition Varchar (12)
CognitiveFunction Varchar (12)
Fall Boolean
HighBloodPressure Boolean
Arrhythmia Boolean
Osteoporosis Boolean
DiabetesMellitus Boolean

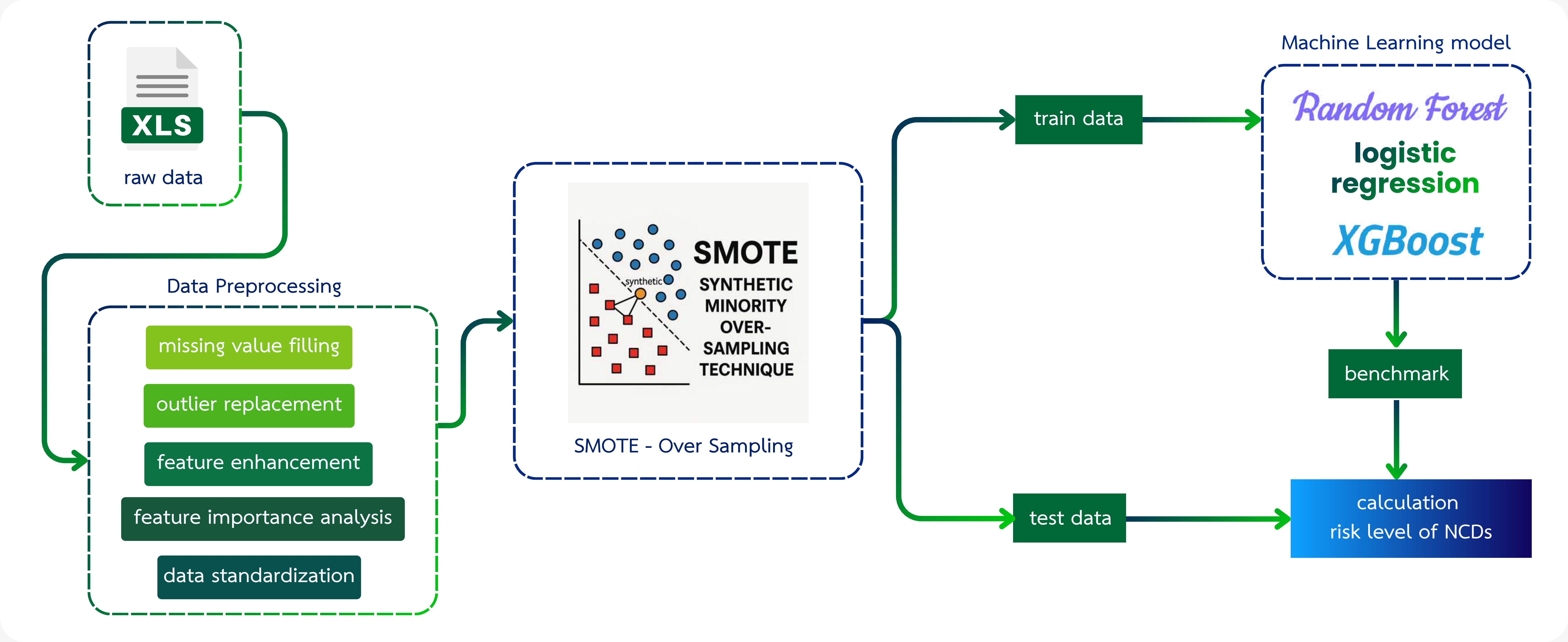
Structural Design

OPERATING ENVIRONMENT

Component	Specification
Operating System	Windows 11 Pro, 64-bit or Linux Ubuntu 22.04 LTS
Database Management System (DBMS)	MySQL 8.0 / PostgreSQL 15 / SQL Server 2022
Hardware Requirement	CPU 4-core, RAM 8GB, Storage 100GB
Application Language / Framework	Python 3.11, Django 4.2 / Java 17, Spring Boot 3.2
Browser / Client	Chrome 120+, Edge 120+

Structural Design

METHODOLOGY



Structural Design

PYTHON LIBRARY

pandas

Data manipulation & analysis

NumPy

Numerical computing with arrays

scikit learn

Machine learning & model evaluation

matplotlib

Data visualization & plotting

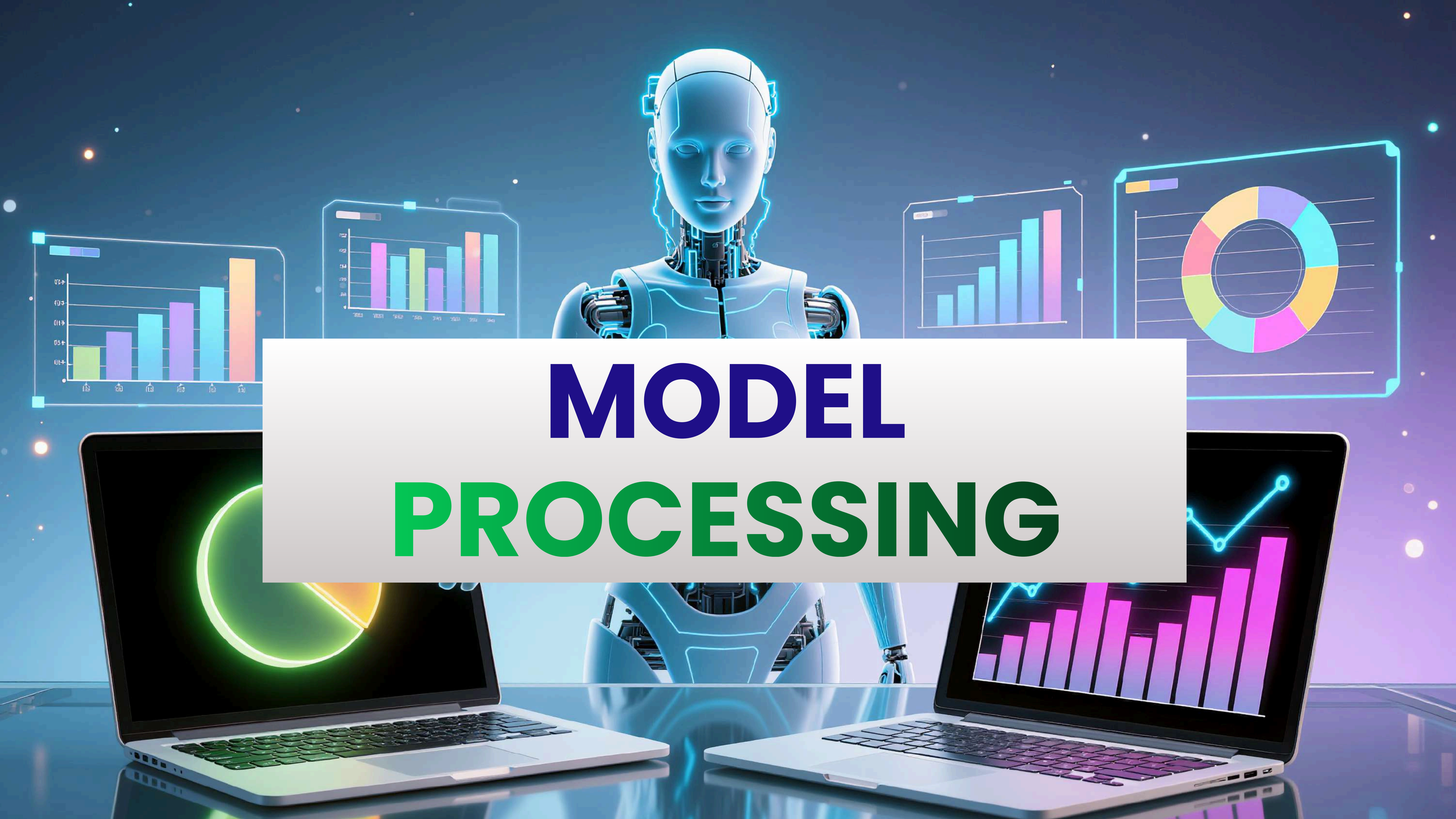
seaborn

Stylish data visualization

Structural Design

TECH STACK

Python IDE	AI APIs	FULL-STACK	DATABASE	DEPLOYMENT	UI/UX DESIGN
Google Colab	FastAPI	Next.JS	PostgreSQL	Docker	Figma
		Tailwind	MongoDB	DigitalOcean	


MODEL
PROCESSING

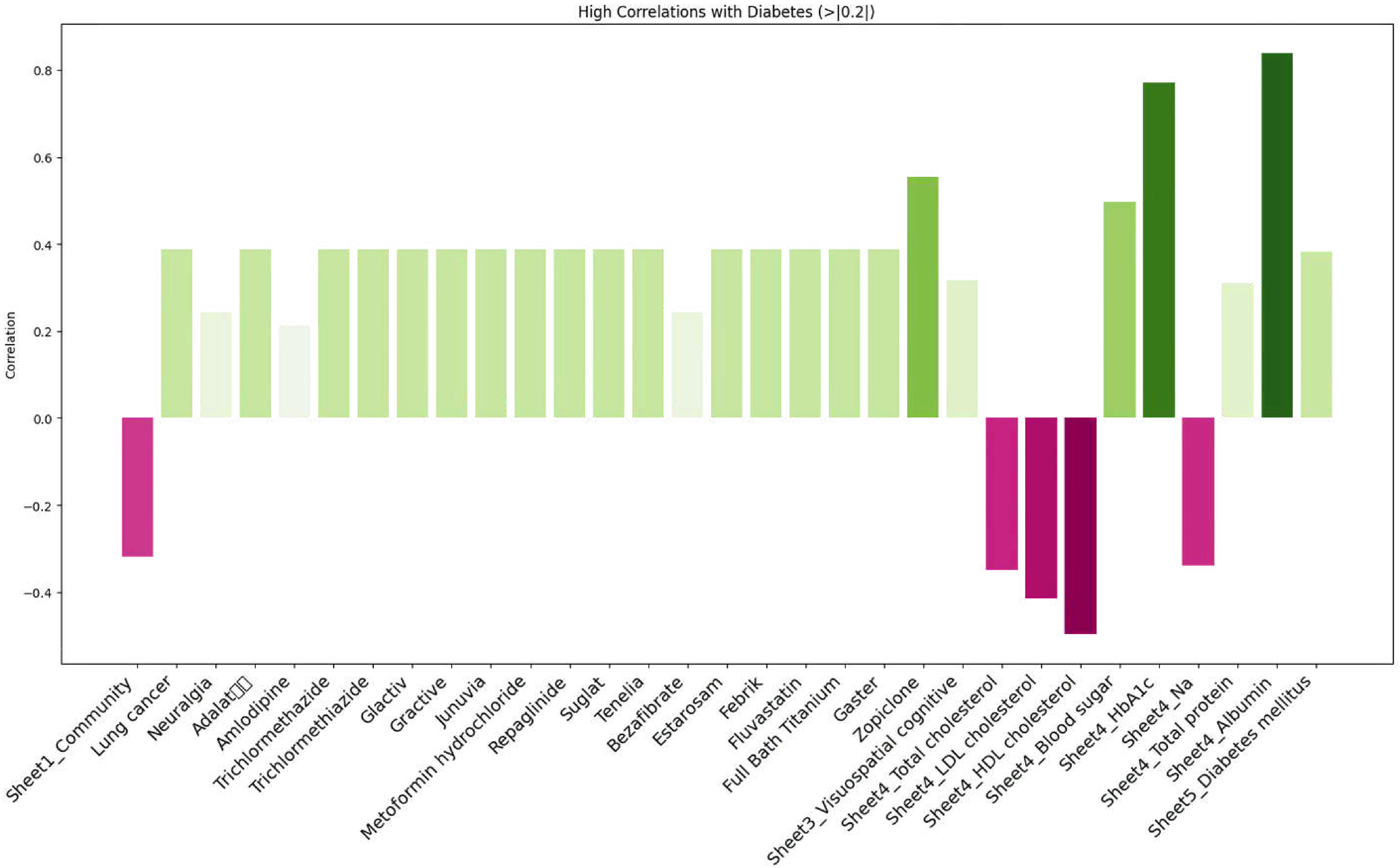
Model Processing

EDA


Correlation with NCD Targets
Sheet1_House Type
Sheet1_Number of residents
Sheet1_Stride（cm)
Sheet1_Drinking
Sheet1_Community
Sheet1_Whether or not to use a pedometer on a daily basis
Sheet1_BMI
Diabetes
Hypercholesterolemia
Lung cancer
Osteoarthritis of both knees (after surgery)
cerebral infarction
Amlodipine
Calblock
Nifedipine
Pisoprolol fumarate
Trichlormethiazide
Glactiv
Metformin
Suglat
Atorvastatin
Crestor
Etizolam
Fluvastatin
Halcyon
Lipitor
Magnesium Oxide
Montelukast
Rivalo
Silodosin
Tumura Kampo Kakkonto
Sheet2_Eating habit
Sheet2_Anxiety about health
Sheet2_life
Sheet3_Remember
Sheet3_Calculation
Sheet4_LDL cholesterol
Sheet4_Creatinine
Sheet4_HbA1c
Sheet4_LDH
Sheet4_ECG abnormalities
Sheet4_Albumin
Sheet5_Exercise function
Sheet5_Fall
Sheet5_Osteoporosis
Sheet5_Nutrition guidance
High blood pressure
Diabetes
Hypercholesterolemia
Hyperlipidemia
Kidney disease
during stent insertion)
during chemotherapy)
Lung cancer
knees (after surgery)
Neuralgia
Uterine fibroid
arrhythmia
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75

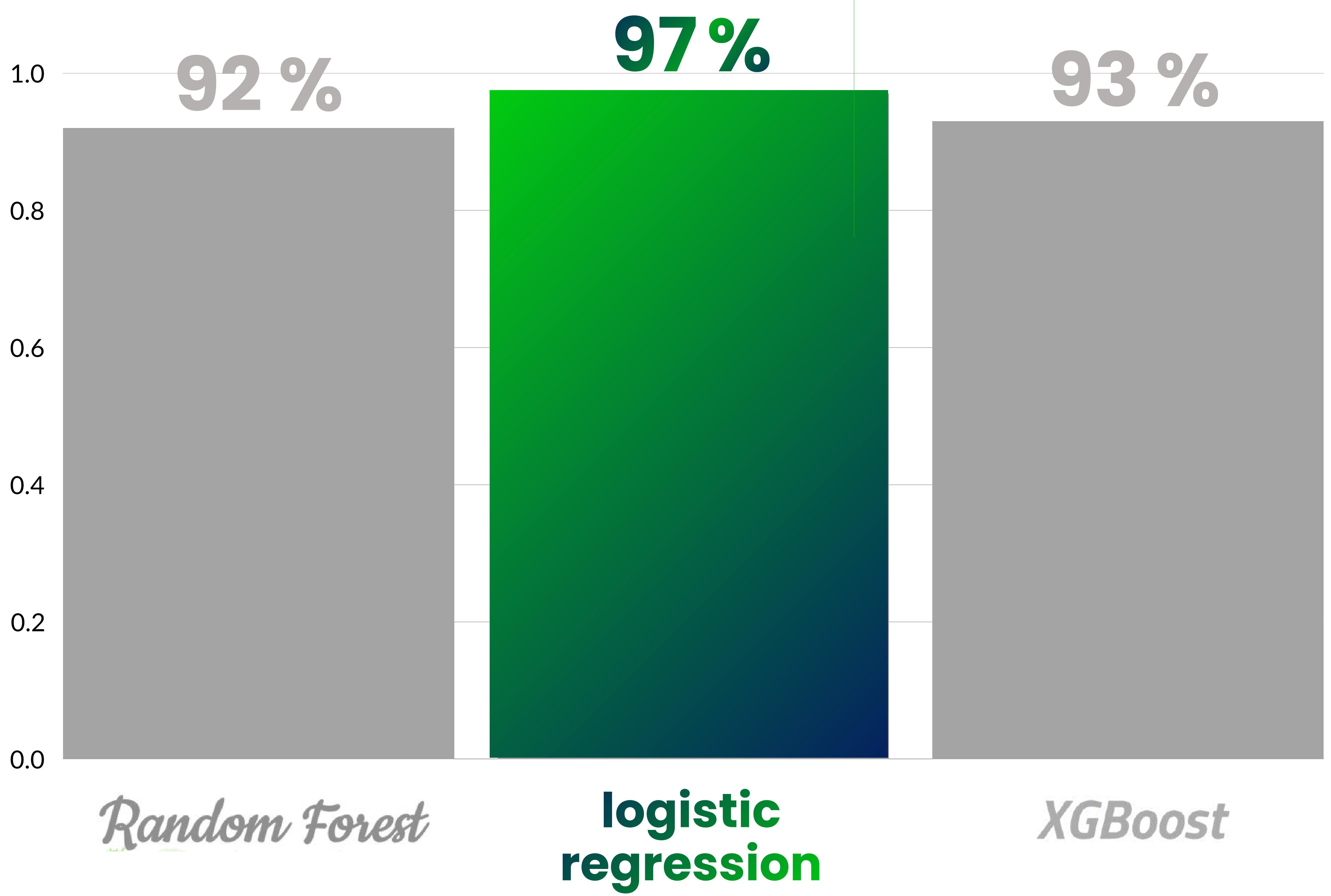
Model Processing

EDA



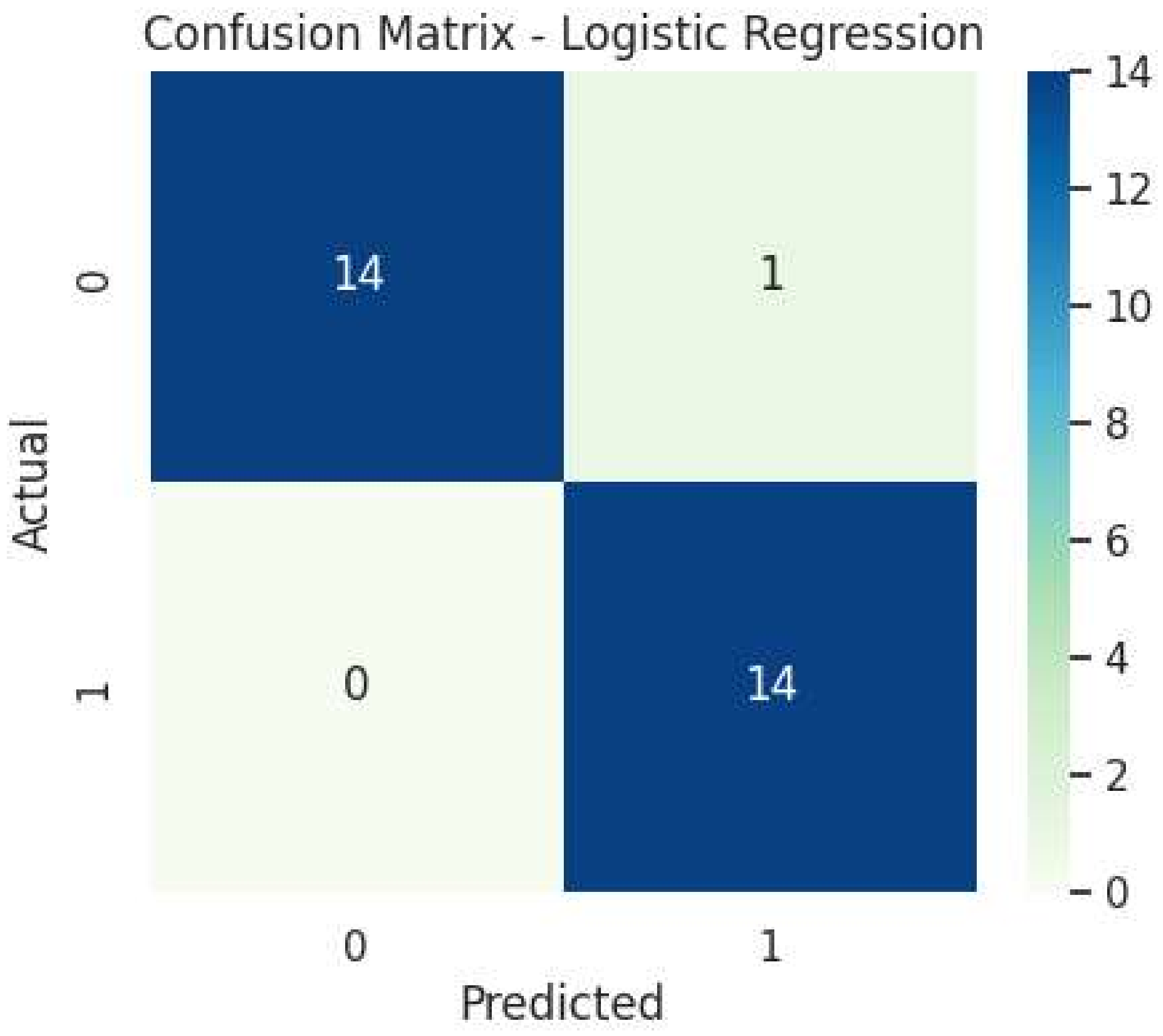
Model Processing

MODEL



Model Processing

MODEL



Accuracy

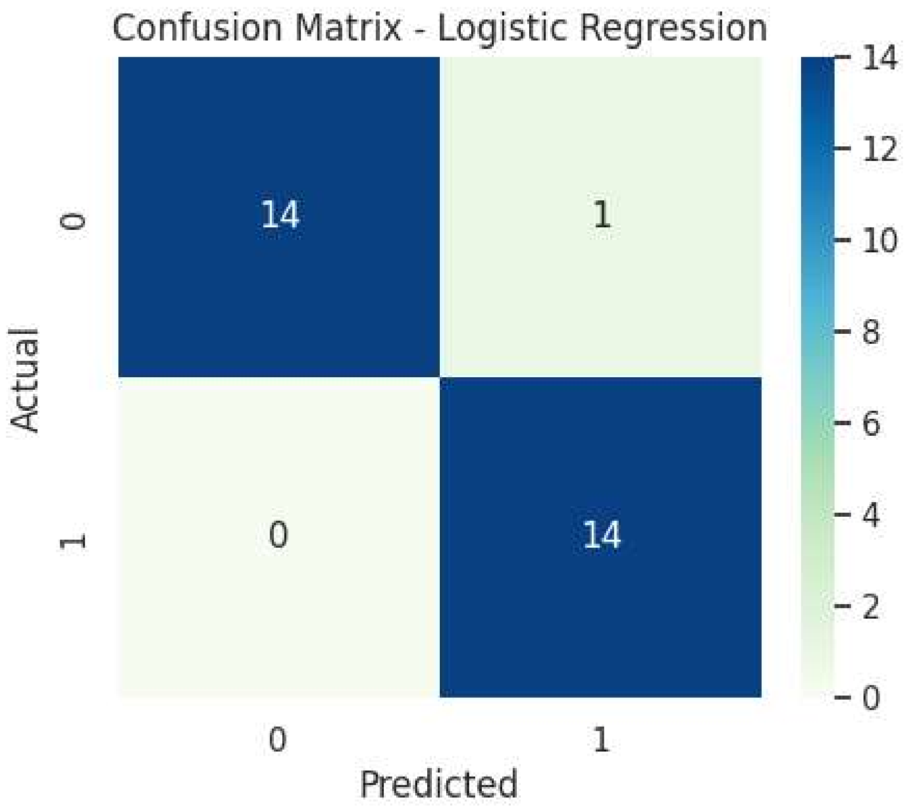
97 %

classification	precision	recall	f1-score
No diabetes	1.00	0.93	0.97
Has diabetes	0.93	1.00	0.97

Model Processing

MODEL

29 EXAMPLES



Probability (%)	Risk Level
6.832433047088343	Low Risk
0.10560155971731891	Low Risk
96.69672996294202	High Risk
96.11832544016843	High Risk
11.46982037825066	Low Risk
94.19146821907687	High Risk
0.6902525295412796	Low Risk
1.7628915817386137	Low Risk
60.417125817382946	Medium Risk
98.94115351479951	High Risk
7.475477895893771	Low Risk
84.43340580614482	High Risk
96.26596141527172	High Risk
2.835318298724958	Low Risk
95.04376362210995	High Risk
96.3541913902479	High Risk
0.24411306336699226	Low Risk
18.082641645375297	Low Risk
4.476662088987894	Low Risk
99.33108264504352	High Risk
0.836487723127201	Low Risk
1.2158931934283352	Low Risk
95.26557233405363	High Risk
2.393070115023066	Low Risk
99.7564525406468	High Risk
96.48448830964765	High Risk
35.89753583188518	Medium Risk
82.2236887480989	High Risk
60.141802845174674	Medium Risk

Range	Risk
0.00 – 0.29	Low Risk
0.30 – 0.69	Medium Risk
0.70 – 1.00	High Risk

UNIT TEST
& OPERATION TEST

Unit Test & Operation Test

UNIT TEST

Activity Level

Stay room ➡ Low
Join / Hobby ➡ Medium
Hard work ➡ High

input_activity	activity_level_output
Work (5 days / week)	High
Housing complex manager	High
teacher of Karaoke	Medium
Handmade class (every day)	Medium
Stay in the room all day	Low
0	Low

Diabetes Risk Level

0.00 – 0.29 ➡ Low
0.30 – 0.69 ➡ Medium
0.70 – 1.00 ➡ High

prob	risk_level_output
0	Low Risk
0.1	Low Risk
0.29	Low Risk
0.3	Medium Risk
0.55	Medium Risk
0.69	Medium Risk
0.7	High Risk
0.95	High Risk

BMI Level

< 18.5 ➡ Underweight
18.5 – 24.9 ➡ Normal
25 – 29.9 ➡ Overweight
30 + ➡ Obese

weight	BMI_Level
17.0	Underweight
18.4	Underweight
18.5	Normal
23.0	Normal
24.9	Normal
25	Overweight
28.5	Overweight
29.9	Overweight
30	Obese
32	Obese

Unit Test & Operation Test

OPERATION TEST

Test ID	Test Scenario	Test Steps	Expected Result	Status
OT01	Open dashboard page	1. Log in as doctor 2. Navigate to dashboard	Dashboard loads successfully showing patient list, selection dropdown, and latest data	T
OT02	Select a doctor	1. Select a doctor from “Select Doctor” dropdown 2. Verify UI updates	Patients list updates to show that doctor’s patients	T
OT03	View patient details	1. Click on a patient record 2. Display patient info and vitals	Patient info page shows age, gender, last visit, vitals, etc.	T
OT04	Verify data display	Browse patient list	All fields (ID, BP, HR, Temp, Weight) are visible and correctly formatted	T
OT05	Search for a patient	1. Enter partial/full name in search box 2. Execute search	List filters to matching patient(s)	T
OT06	Navigation consistency	Use breadcrumb/back buttons to return to list	Page returns without losing context	T

OUR TEAM

7202

Kankrit Thamwirotsiri

7209

Thanuwach Thanachoksirirat

7211

Taechapat Akejiratrakul

7233

Phumin Boontrarickamas

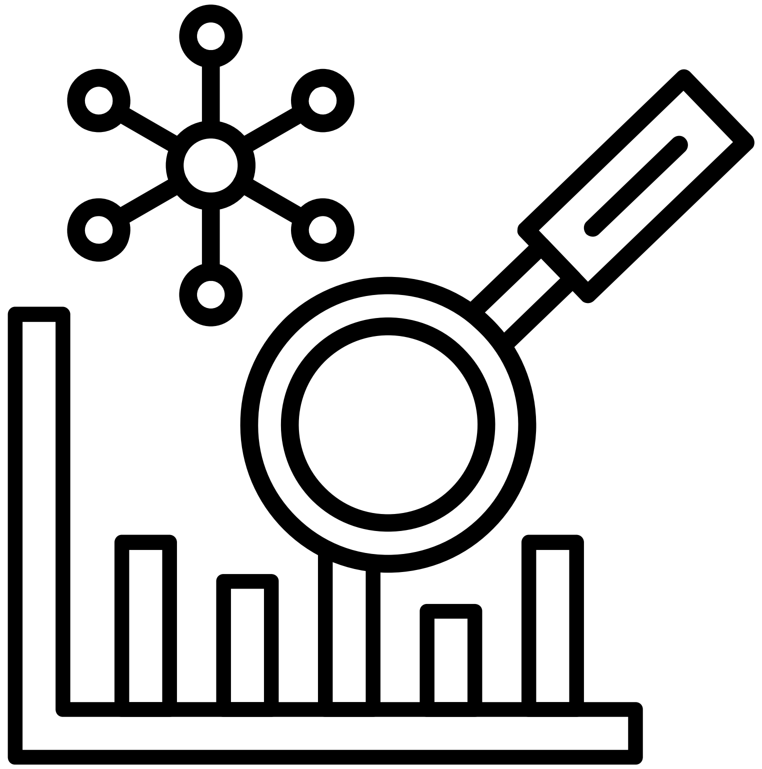
7236

Punchaya Chancharoen

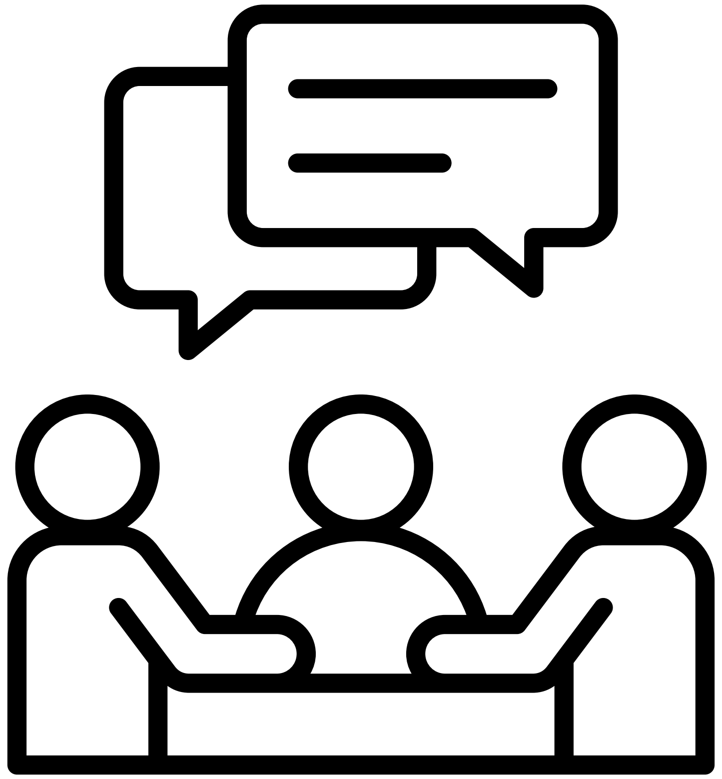

DEMO WEBSITE FOR PC
LINK HERE

Appendix

CHALLENGE PROBLEM



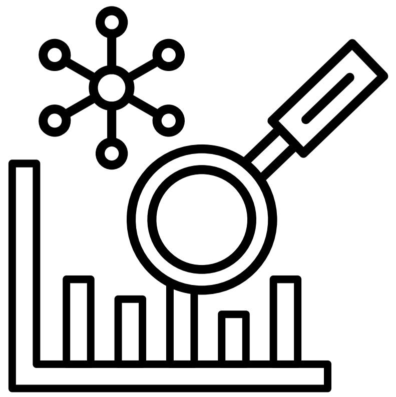
Complex and messy data



online meeting and working

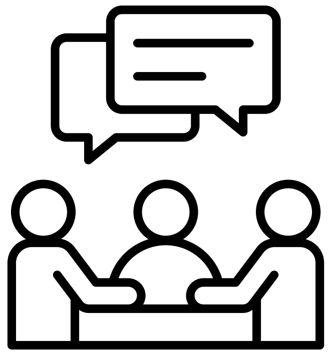
Appendix

CHALLENGE SLOVING



Complex and messy data

- Gained a deep understanding of the data structure and context
- Cleaned and transformed the data to make it easier to read, analyze, and manage
- Organized data systematically to support accurate analysis and decision-making



Online meeting and working

- Communicated frequently and consistently
- Created an open space for team members to freely express opinions and ideas
- Fostered a positive and collaborative working environment

Appendix

NEXT STEPS FOR IMPLEMENTATION

Data Quality

- Data is complete, accurate, and up to date
- Missing values and noise are minimized

Model & Analytics

- Predictions are accurate and reliable
- Results are easy to understand

Data Privacy

- Personal data is protected
- Data security measures are in place

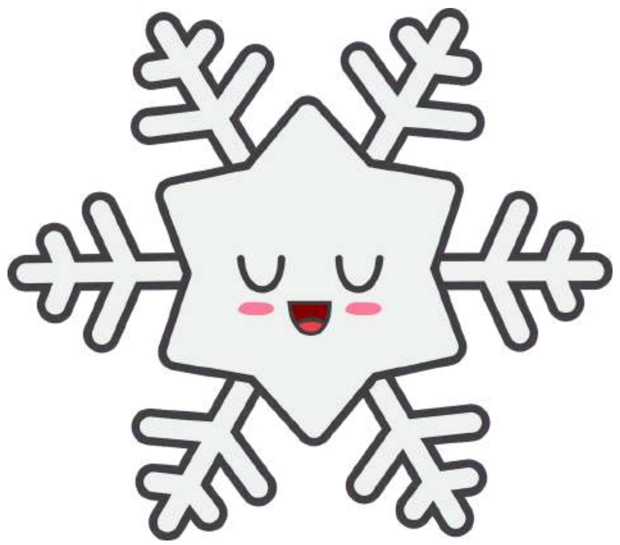
Appendix

DIABETES RISK LEVEL

< 30% **Low Risk**	**HbA1c** < 5.7% **BMI** < 23
30 – 70% **Medium Risk**	**HbA1c** : 5.7% – 6.4% **BMI** > 23
> 70% **High Risk**	**HbA1c** > 6.5% **Fasting Blood Sugar** =126 mg/dL

Appendix

FUNCTION



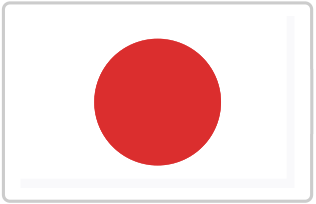
Diabetes Risk Level

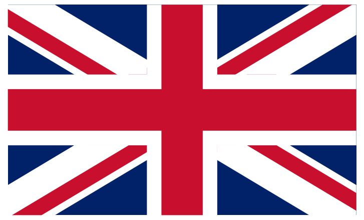
0.00 – 0.29	Low
0.30 – 0.69	Medium
0.70 – 1.00	High

BMI Level

< 18.5	Underweight
18.5 – 24.9	Normal
25 – 29.9	Overweight
30 +	Obese

Clean Text





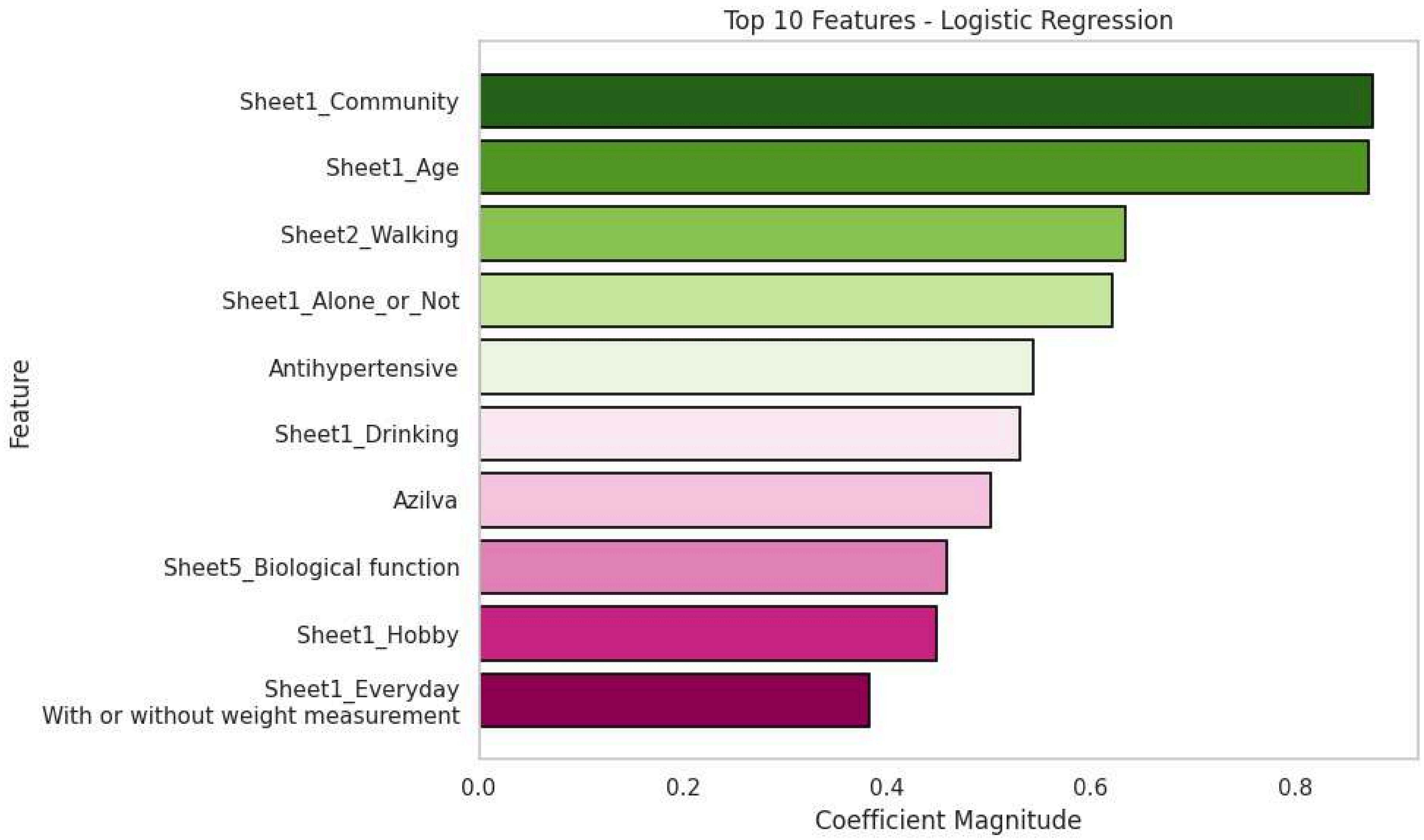
Diabetes mellitus	Diabetes
List Columns	Single Columns
Complex text	Simple text

Activity Level

Stay room	Low
Join / Hobby	Medium
Hard work	High

Appendix

CORRERATION'S MODEL



Appendix

ANOTHER PREDICTION OF NCDS

ตารางที่ 1. ปัจจัยเสี่ยงของโรคเบาหวานชนิดที่ 2 และคะแนนความเสี่ยง[2]

ปัจจัยเสี่ยง	คะแนนความเสี่ยง Diabetes risk score
อายุ	
• 34 – 39 ปี	0
• 40 – 44 ปี	0
• 45 – 49 ปี	1
• ตั้งแต่ 50 ปีขึ้นไป	2
เพศ	
• หญิง	0
• ชาย	2
ดัชนีมวลกาย	
• ต่ำกว่า 23 กก./ม.^{2}	0
• ตั้งแต่ 23 ขึ้นไปแต่ น้อยกว่า 27.5 กก/ม.^{2}	3
• ตั้งแต่ 27.5 กก./ม^{2}ขึ้นไป	5
รอบเอว	
• ผู้ชายน้อยกว่า 90 ซม. ผู้หญิงน้อยกว่า 80 ซม.	0
• ผู้ชายตั้งแต่ 90 ซม. ขึ้นไป, ผู้หญิงตั้งแต่ 80 ซม. ขึ้นไป	2
ความดันโลหิต	
• ไม่มี	0
• มี	2
ประวัติโรคเบาหวานในญาติสายตรง (พ่อ แม่ พี่ หรือ น้อง)	
• ไม่มี	0
• มี	4

อายุ

- ○ 34-39 ปี
- ○ 40-44 ปี
- ○ 45-49 ปี
- ○ 50 ปีขึ้นไป

เพศ

- ○ หญิง
- ○ ชาย

ดัชนีมวลกาย(BMI)

- ○ ต่ำกว่า 23 กก./ม.2
- ○ ตั้งแต่ 23 ขึ้นไปแต่ น้อยกว่า 27.5 กก/ม.2
- ○ มากกว่า 27.5 กก./ม2 ขึ้นไป

รอบเอว(เซนติเมตร)

- ○ ผู้ชายน้อยกว่า 90 ซม. ผู้หญิงน้อยกว่า 80 ซม.
- ○ ผู้ชายตั้งแต่ 90 ซม. ขึ้นไป, ผู้หญิงตั้งแต่ 80 ซม. ขึ้นไป

ความดันโลหิต

- ○ ไม่มี
- ○ มี

ประวัติโรคเบาหวานในญาติสายตรง (พ่อ แม่ พี่ หรือ น้อง)

- ○ ไม่มี
- ○ มี

Siriraj Diabetes Center of Excellence, Faculty of Medicine Siriraj Hospital

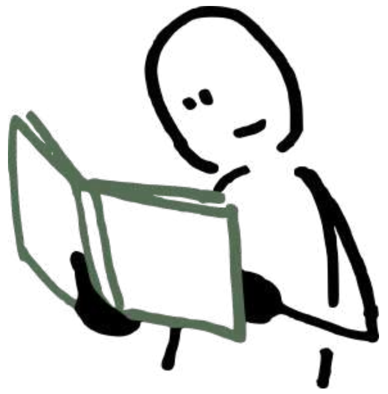
Appendix

ACTIVITY LEVEL

STAY HOME

Low activity

0.0 –1.9



1.0–1.3 METS

1.3–1.5 METS

HOBBY

Medium activity

2.0 – 5.9

3–4 METS

2.5–3.5 METS

WORK HARD

High activity

≥6.0

6–10 METS

ref : METs (Metabolic Equivalent of Task)

Appendix

DIABETES RISK LEVEL

HbA1c (Hemoglobin A1c)

- **Normal:** Below 5.7%
- **Pre-diabetes:** 5.7% – 6.4%
- **Diabetes:** 6.5% or higher

Fasting Blood Sugar (FBS)

- **Normal:** 70 – 99 mg/dL
- **Pre-diabetes:** 100 – 125 mg/dL
- **Diabetes:** 126 mg/dL or higher

BMI (Body Mass Index) - Asian Criteria

- **Normal:** 18.5 – 22.9 kg/m²
- **Overweight:** 23.0 – 24.9 kg/m²
- **Obese:** 25.0 kg/m² or higher